MACHINE LEARNING AND RELATED APPLICATIONS

# Review of Research Study

# Air-quality prediction based on the ARIMA-CNN-LSTM combination model optimized by dung beetle optimizer

## (Phase 1)

K.W.G.C.D.B. Karunarathne

KIC- BSCCSAI-251P-036

Kandy Innovation Centre,

National Institute of Business Management,

## Introduction

This report reviews Duan et al. (2023), who forecast the daily Air Quality Index (AQI) of four Chinese cities using an ARIMA–CNN–LSTM hybrid whose hyperparameters are tuned by the Dung Beetle Optimizer (DBO). Phase 1 deconstructs the study along a single machine-learning pipeline: data preprocessing → ML training → ML optimiser → evaluation.

Section (a) formulates the problem and records the errors found in the paper. Section (b) justifies the methodology, presents the workflow and examines the mathematics of training and optimisation. Section (c) analyses the dataset, the preprocessing pipeline and the evaluation method. Phase 1 supplies the problem definition, the workflow and the list of weaknesses that Phase 2 (baseline implementation) will test and Phase 3 (critical review) will assess. Notation follows the paper: $x_t$ is the AQI on day t and $\hat{y}_t$ is the forecast.

## (a) Problem Formulation

In standard machine-learning terminology, the paper formulates air-quality forecasting as a supervised, univariate time-series regression problem: given a sequence of past observations of an outcome variable, the model learns a function that maps a fixed-length window of historical values to a real-valued prediction of a future observation. This is a regression task rather than a classification task because the output — the Air Quality Index (AQI) — is a continuous numerical quantity, not a discrete category.

Target variable: the AQI value of a given city on a future day. AQI is itself an upstream composite index calculated from six pollutant concentrations (PM2.5, PM10, NO2, SO2, CO, O3), but the authors do not model these six pollutants separately — the pre-computed AQI figure is the single dependent variable being predicted for each of the four study cities (Beijing, Lanzhou, Jiaozuo, Guangzhou).

Input features: the only inputs used are lagged values of the same AQI series, taken over a sliding window of 10 time-steps (Table 2). No exogenous variables — weather, traffic, season, or the individual pollutant concentrations that make up AQI — are included as features, even though several papers cited in the authors' own literature review (e.g. Rakholia et al.; Jurado) do incorporate such variables. For the proposed model specifically, the input window is further decomposed by the ARIMA model into a linear component and a residual (non-linear) component, and only the residual is passed to the CNN-LSTM network as its effective input.

Learning paradigm: this is a supervised learning problem. Ground-truth AQI values are available for every time-step, and every model (ARIMA, RF, SVM, LSTM and the various hybrids) is trained by minimising the error between its predictions and these known, labelled values, then judged on a held-out test split using RMSE, MAE and $R^2$. It is not an unsupervised problem, since no clustering or density-estimation objective is used on unlabelled data. It is also not reinforcement learning: there is no agent, no environment to interact with, no sequential decision-making, and no reward signal being maximised over time. This point is worth stating explicitly because the paper's use of the Dung Beetle Optimizer — a bio-inspired "foraging/rolling" search metaheuristic — could be mistaken for a reinforcement-learning component; in fact, DBO is used purely as an offline hyperparameter-search algorithm for the supervised CNN-LSTM network, not as a learning paradigm in its own right.

**Real-world context and users.** The model supports short-term air-quality forecasting for environmental agencies and the public, who use predicted AQI to issue health advisories and plan

emission-control measures. The decision supported is usually threshold-based (for example, whether tomorrow's AQI will pass an unhealthy level), so accuracy on high-AQI days matters more than average accuracy.

**Task and scope.** The task is one-step-ahead univariate time-series regression, repeated independently for four cities (Beijing, Lanzhou, Jiaozuo and Guangzhou). The forecast horizon is not stated explicitly; a one-step-ahead forecast is implied by the sliding-window setup (window = 10, Table 2). Scope limits are that only the AQI history is used, only four Chinese cities are studied, and the authors themselves note that meteorological and seasonal factors are not considered.

**Assumptions.** (i) The series becomes stationary after first differencing (d = 1 in Table 1). (ii) The linear and non-linear parts are additive, $x_t = L_t + N_t$ (Eq. 21). (iii) The 2015–2022 pattern stays representative in the test period, which policy changes and COVID-19 may violate. (iv) The observations are complete and reliable.

**What a good and a bad result means.** The paper reports RMSE, MAE and $R^2$. A good result is a low error in AQI units that also beats a naive persistence forecast ($\hat{y}_t = x_{t-1}$) and a plain ARIMA over the whole test period, including peak days. A bad result only matches persistence, or shows a sharp rise in error when AQI exceeds about 150. $R^2$ close to 1 on a trending, seasonal series is not by itself evidence of skill, because the variance in its denominator is inflated by that structure. The paper reports no persistence or mean-predictor baseline, so the size of the claimed improvement cannot be judged.

## Errors and inconsistencies found in the paper

| Location in paper | Problem found | Consequence |
|---|---|---|
| Eq. (25) | $R^2$ is printed as $\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2$ with no "1 −", which is $1 - R^2$. | Metric defined incorrectly as written |
| Eqs. (22)–(23) | The min-max denominator is printed as "xminmax" instead of x_max − x_min; the inverse transform is garbled in the same way. | Scaling cannot be reproduced from the text |
| Text under Eq. (2) vs Table 1 | The text says ARMA(p, 2, q) is needed (second-order differencing), but every city uses d = 1. | Inconsistent differencing order |
| Eq. (1) | $y_t$ is described as "the number of difference levels"; it is the (differenced) series value. | Wrong variable definition |
| Eq. (4) | The forget gate uses a dot product $W_f \cdot [h_{t-1} \cdot x_t]$, while Eqs. (5)–(8) use concatenation $[h_{t-1}, x_t]$. | Notation error |
| DBO text, Eqs. (10), (12) | $x_i(t)$ is called the position of the "praying mantis"; Eq. (12) uses (1 − R) for both bounds while Eq. (14) uses (1 + R) for the upper bound. | Copy errors; Eq. (12) appears wrong |
| Discussion | Jiaozuo is called the most accurate city, but Guangzhou has lower RMSE (5.496 vs 7.594) and MAE (3.77 vs 5.285); only $R^2$ favours Jiaozuo. | Claim not supported by the numbers |
| Fig. 8 vs Fig. 9 | The RMSE bars for Lanzhou are identical to Beijing's (64.617 … 14.94) although MAE and $R^2$ differ; Lanzhou's MAE of 5.12 sits beside an RMSE bar of 14.94, while the abstract lists 7.841. | Probable copy-paste error in Fig. 9 |
| Results, point D | A 64.02 % RMSE reduction is claimed versus CEEMDAN-CNN-LSTM, but Fig. 7a (35.002 → 7.594) gives about 78.3 %; "84.78 %" is also repeated twice. | Arithmetic inconsistent with the figure |

| Location in paper | Problem found | Consequence |
|---|---|---|
| Results, point C | $R^2$ is said to have "decreased" by 39.72 %, but Fig. 7a shows it rose (0.506 → 0.707). | Wording contradicts the data |
| Figs. 7–10 | The models labelled A–J are never defined in a caption or legend. | Bars cannot be matched to models with certainty |
| Table 2 | No hyperparameters are given for Random Forest, and SVM uses "other parameters default". | Baselines not reproducible |
| Study area, Results | The text switches between "four cities" and "these three cities". | Inconsistent scope |

## Criticism of the dataset

**Sampling bias.** Four industrial cities were chosen because they are heavily polluted, so the results may not generalise to cleaner or non-Chinese cities. Guangzhou's AQI (about 0–200 in Fig. 5) behaves very differently from the other three (up to about 500).

**Access and reproducibility.** The data are licensed and not public (Data availability statement), so no one can independently check the results.

**Measurement and definition.** AQI is a derived index that is capped near 500, and the paper does not say which AQI standard or station aggregation was used, or whether any days are missing.

**Distribution shift.** The period 2015–2022 contains policy interventions and COVID-19. Fig. 7's time axis suggests the test set starts near day 2080, roughly the last 18 months, which is a single narrow window.

## Ethical points

Privacy risk is low because the data are city-level aggregates. The main concerns are misuse and over-trust: an overclaimed $R^2$ could lead agencies or the public to rely on forecasts that fail on exactly the high-pollution days that matter. Results from four industrial Chinese cities should also not be applied to other regions without re-validation.

| Attribute | Detail | Notes |
|---|---|---|
| Series | Daily AQI, 4 cities: Beijing, Lanzhou, Jiaozuo, Guangzhou | Treated as 4 independent univariate experiments |
| Period / Size | Jan 2015 – 1 Mar 2022 (~2,600 daily points per city) | ~7 years of continuous daily data |
| Source | Resource and Environment Science and Data Center, Chinese Academy of Sciences | Secondary/administrative data, not authors' own sensors |
| Variables | A single composite index (AQI) derived upstream from PM2.5, PM10, NO2, SO2, CO, O3 | The six underlying pollutants are not used individually as features |

## (b) Methodological Justification

Figure 1 shows the workflow as implemented by the authors. The first step is data preprocessing, and the red boxes mark steps the paper does not describe or leaves ambiguous.

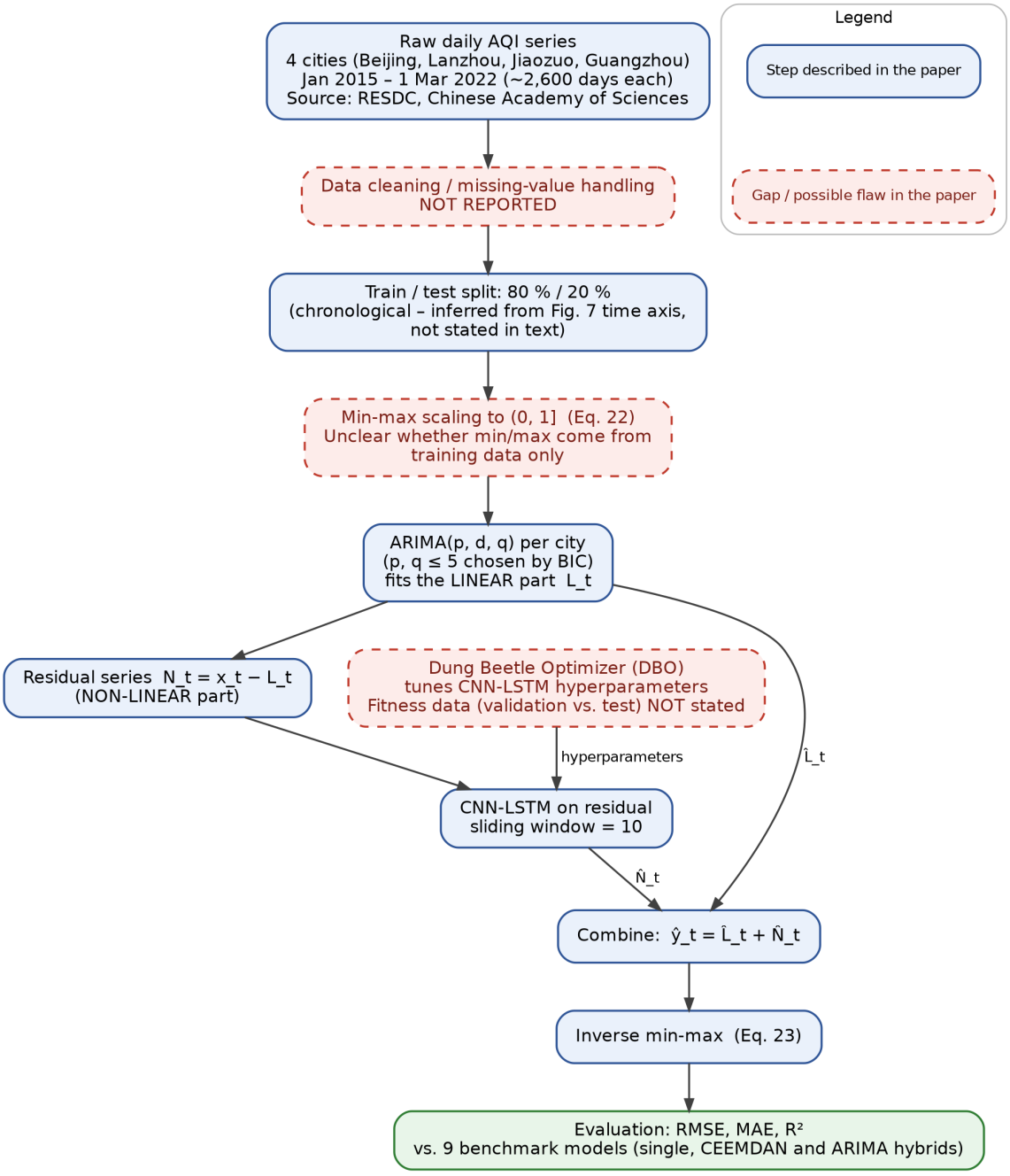



*Figure 1: Workflow of the reviewed method (red dashed boxes = gaps or possible flaws)*

## Why each step is done, the alternatives, and the verdict

| Step | Why it is done | Alternatives | Verdict on the authors' choice |
|---|---|---|---|
| Cleaning | Give the model a complete, valid series. | Interpolation, forward-fill, dropping, model-based imputation. | Not described, so it cannot be judged. |
| Train/test split (80/20) | Estimate performance on unseen data. | Random split (invalid for time series), train/validation/test, walk-forward validation. | A chronological hold-out is the right family, but a single split with no validation set is weak. |
| Min-max scaling | Neural networks and SVMs are scale-sensitive; the LSTM gates (sigmoid/tanh) work best on small inputs. ARIMA and Random Forest do not need scaling. | Standardisation (z-score), robust scaling, log transform. | Acceptable, but one spike near 500 compresses all other values, and test values above the training maximum leave (0, 1]. Scaler must be fitted on training data only. |
| ARIMA for the linear part | Captures autocorrelation and removes trend by differencing. | SARIMA, exponential smoothing, STL, CEEMDAN. | Reasonable, but no seasonal term although Fig. 5 shows yearly cycles. |
| CNN-LSTM for the non-linear part | CNN extracts local patterns, LSTM models long dependencies. | GRU, temporal CNN, Transformer, attention-LSTM, gradient boosting on lag features. | Plausible but not compared with any other deep architecture. |
| DBO tuning | Replaces manual ("blind") hyperparameter choice. | Grid, random or Bayesian search; GWO, PSO. | Not compared with any alternative optimiser; fitness data not stated. |
| RMSE, MAE, $R^2$ | Measure error size and explained variance. | MAPE, MASE, persistence baseline, walk-forward validation, peak-day metrics. | Standard, but incomplete for forecasting (see section on evaluation). |

No feature selection or extraction such as PCA is used: with a single input variable there is nothing to select. The CEEMDAN step in the comparison models does the opposite, expanding one series into 9–10 components, which is why it must be fitted without future data.

The proposed ARIMA-DBO-CNN-LSTM pipeline combines three separate techniques, each targeting a different weakness the authors identify in single-model approaches.

- ARIMA for the linear component – the authors justify this by noting that AQI series contain a short-period linear trend that classical linear models capture reliably, whereas purely non-linear deep-learning models can "blind" their own hyperparameter search if asked to fit both the linear and non-linear structure at once. ARIMA order (p, d, q) is chosen per city using the Bayesian Information Criterion, capped at $p, q \leq 5$, which is a standard, defensible model-selection procedure.
- CNN-LSTM for the non-linear component – LSTM is chosen over a plain recurrent network because its forget/input/output gate structure avoids the vanishing-gradient problem that limits how far back a simple RNN can "remember," which matters for a ~2,600-point daily series. The CNN layer is added in front of the LSTM to automatically extract local temporal patterns before the LSTM models longer-range dependencies. This choice is justified by AQI

data being “non-constant and non-linear,” a property linear models such as ARIMA cannot capture on their own.

- Dung Beetle Optimizer (DBO) for hyperparameter tuning – the authors state that CNN-LSTM hyperparameters (neuron counts, filter size, kernel size, stride, learning rate, batch size, sliding window) are normally set “blindly” by manual trial and error, and use DBO, a 2022 population-based metaheuristic, to search this space automatically instead.

Why these were chosen over alternatives: the paper's justification is comparative in one respect and purely descriptive in another. It is comparative for the overall architecture — the authors run an ablation-style sequence of ten models (single models through to the full hybrid) and show a consistent, monotonic improvement in RMSE, MAE and $R^2$ as each component (ARIMA decomposition, CNN layer, DBO tuning) is added. This is a reasonably strong empirical argument for combining these three particular techniques rather than using any one of them alone.

However, the justification is weaker when it comes to explaining why these specific techniques were chosen over comparable alternatives at each stage. The authors' own literature review lists several optimizers already applied to the same AQI-forecasting problem — Grey Wolf Optimizer, Particle Swarm Optimization, Harris Hawks Optimization, and Cuckoo Search — yet DBO is adopted without being run head-to-head against any of them within this study; its selection rests on general claims of “strong merit-seeking ability and fast convergence” rather than a comparative experiment. Likewise, no seasonal variant of ARIMA (e.g. SARIMA) is considered, despite the raw AQI plots (Fig. 5) showing visible annual/seasonal cycles, particularly in Beijing, Lanzhou and Jiaozuo. On the deep-learning side, the paper does not test attention-based or Transformer-style alternatives, even though an attention-augmented LSTM is cited approvingly in the introduction (Wang et al.) as improving prediction accuracy. As a result, the paper convincingly shows that combining ARIMA, CNN-LSTM and DBO outperforms not combining them, but it does not fully establish that this particular combination is superior to every reasonable alternative combination.

# Mathematics of training and the optimiser

## Mathematical aspects of training and optimisation

**Loss and optimiser.** The paper states neither the loss function nor the gradient optimiser. For regression the natural choice is the mean squared error, $L = (1/n)\,\Sigma(\hat{y}_i - y_i)^2$, minimised by mini-batch gradient descent, most commonly Adam (Kingma & Ba, 2015). These must be assumed, which is itself an omission. Table 2 gives batch sizes of 64–100, 100 epochs and learning rates of 0.1–0.2, which are very high for an LSTM (1e−3 is typical) and risk unstable training. No dropout, L1/L2 penalty or early stopping is mentioned.

**ARIMA (statistical part).** The model is

$$y_t = c + \varphi_1 y_{t-1} + \ldots + \varphi_p y_{t-p} + \theta_1 e_{t-1} + \ldots + \theta_q\, e_{t-_q} \quad \text{(Eq. 1)}$$

$$\mathrm{BIC} = k \ln(n) - 2 \ln(L) \quad \text{(Eq. 2)}$$

where $y_t$ is the d-times differenced series, $\varphi$ and $\theta$ are estimated by maximum likelihood, k is the number of parameters, n the sample size and L the likelihood. BIC penalises complexity more strongly than AIC (k ln n versus 2k), so it favours small orders; this suits a series of about 2,000 points, and the chosen orders (3,1,1) and (2,1,2) are indeed small.

**CNN-LSTM (architecture).** Table 2 uses 512 convolution filters (kernel 2, stride 1) followed by three LSTM layers of about 50 units. A 1-D convolution computes $z_j = f(\Sigma_k w_k x_{j+k} + b)$, so each filter detects a local pattern. The LSTM cell then updates

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$
$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$
$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$
$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$
$$h_t = o_t \odot \tanh(C_t) \quad \text{(Eqs. 4–9)}$$

The additive cell-state update $C_t = f_t \odot C_{t-1} + \ldots$ gives gradients a path that does not shrink at every step during backpropagation through time, which is why LSTM avoids the vanishing-gradient problem of a plain RNN (Hochreiter & Schmidhuber, 1997). However, 512 filters feeding a 50-unit LSTM gives roughly $1.1 \times 10^5$ parameters in the first LSTM layer alone ($4 \times [50 \times (512 + 50) + 50]$), against only about 2,000 training days. This invites overfitting, and the paper reports no regularisation to control it.

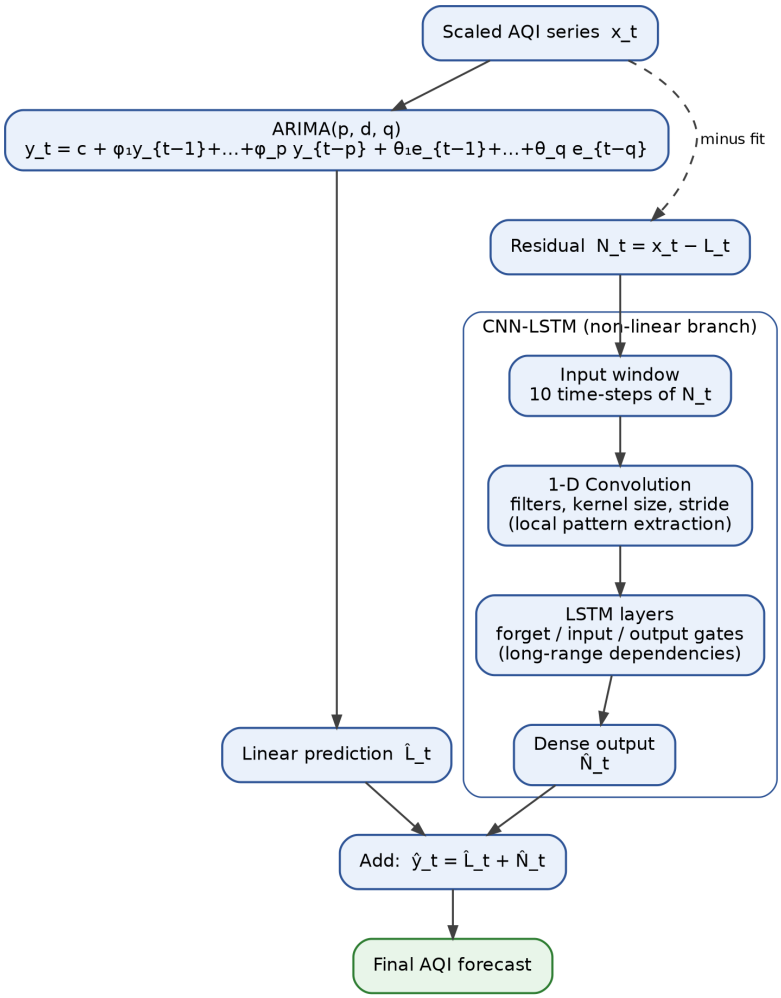



*Figure 3: Structure of the ARIMA + CNN-LSTM hybrid*

**DBO (the optimiser).** DBO is a population-based metaheuristic (Xue & Shen, 2023). Each beetle is a vector of hyperparameters; here the search space in Table 2 has about nine dimensions (three layer

sizes, window, learning rate, batch size, filters, kernel size, stride). Positions are updated by four behaviours: ball rolling $x(t+1) = x(t) + \alpha \cdot k \cdot x(t-1) + b \cdot \Delta x$, dancing $x(t+1) = x(t) + \tan\theta \cdot |x(t) - x(t-1)|$, breeding and foraging inside shrinking bounds, and stealing near the global best. Every fitness evaluation means training a whole CNN-LSTM for 100 epochs, so the cost is (population size × iterations × one training run). The paper reports neither population size, iteration count, convergence curve nor run time. It also does not say how DBO's continuous positions are rounded to integers such as filters, kernel size and stride, nor which data score the fitness. If the test set was used, the final test error is optimistically biased.

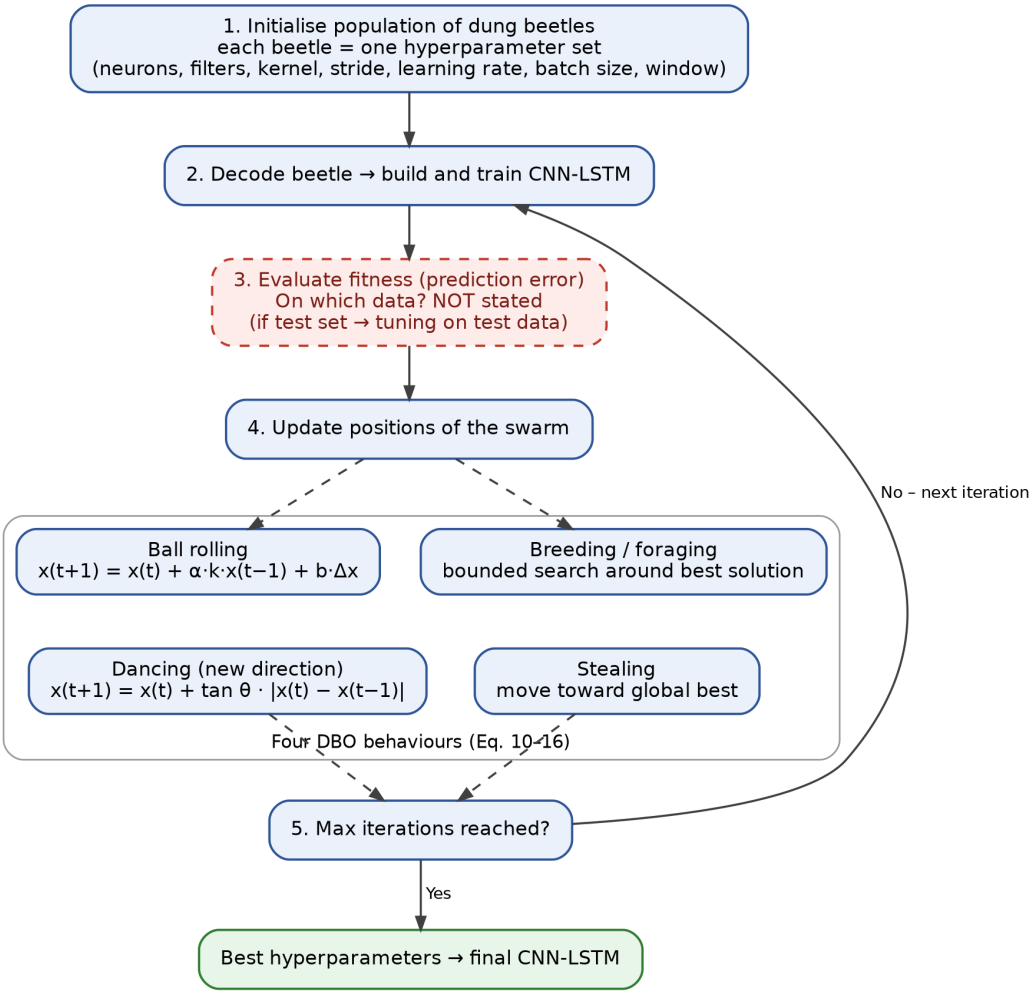



*Figure 4: DBO hyperparameter search loop for the CNN-LSTM*

**Baseline algorithms (SVM and Random Forest).** The paper describes SVM as a binary classifier with margin $2/\|w\|$, but the task is regression, so the correct form is support vector regression: minimise $\frac{1}{2}\|w\|^2 + C\Sigma(\xi_i + \xi_i^*)$ with an ε-insensitive loss. Only a linear kernel $K(x, z) = x \cdot z$ is used, with all other parameters left at defaults. If scikit-learn defaults were used ($C = 1$, $\varepsilon = 0.1$), an ε of 0.1 on data scaled to (0, 1] ignores errors up to 10 % of the AQI range, which would alone explain the weak SVM result. For Random Forest the paper names only the number of trees and number of features, with no values. For regression the split criterion is variance reduction (MSE), not Gini or entropy.

**Bias–variance reading of the results.** The single models reach $R^2$ of only about 0.27–0.5, which suggests high bias or weak baselines, whereas the hybrids reach 0.95–0.99. In Jiaozuo, DBO adds

only 0.016 of $R^2$ (0.973 → 0.989) to ARIMA-CNN-LSTM, so most of the improvement comes from the ARIMA-residual structure rather than from optimisation.

## (c) Data Pipeline Analysis

The dataset consists of daily AQI readings for four Chinese cities, obtained from the Resource and Environment Science and Data Center of the Chinese Academy of Sciences, spanning January 2015 to 1 March 2022 (roughly 2,600 records per city). AQI is a single pre-aggregated numeric feature per day per city; the underlying six pollutant concentrations that feed into AQI are not used directly.

### Statistical analysis and EDA

### Statistical analysis and exploratory data analysis

The paper reports no descriptive statistics, so this section combines what Fig. 5 shows with the tests that should be run. There is one numeric feature per city, so a correlation matrix across features does not apply; the time-series equivalents are the autocorrelation (ACF) and partial autocorrelation (PACF) functions, which would justify the orders p, q and the window of 10.

| City | Range read from Fig. 5 (approx.) | Visual pattern |
|---|---|---|
| Beijing | about 10–480 | Frequent high spikes in 2015–2017, fewer and lower afterwards; strongly right-skewed. |
| Lanzhou | about 30–500 | Stable baseline around 50–120 with a few isolated spikes reaching the 500 ceiling. |
| Jiaozuo | about 20–500 | Clear winter peaks every year; the strongest seasonality of the four. |
| Guangzhou | about 10–200 | Lowest level and narrowest range; spikes up to about 190. |

| Statistic (per city) | Result | What it tells us |
|---|---|---|
| Mean, median, variance, SD | [FILL IN] | Compare mean with median: mean > median confirms right skew. |
| Skewness, kurtosis | [FILL IN] | Expected positive skew and heavy tails because AQI is bounded below by 0 and has rare spikes. |
| Shapiro–Wilk / Q-Q plot | [FILL IN] | Normality is expected to be rejected; a log or Box–Cox transform could then be considered. The paper applies none. |
| ADF stationarity test | [FILL IN] | A unit root would support d = 1; seasonality still needs a seasonal term. |
| Missing days, duplicates | [FILL IN] | Needed to judge the cleaning step the paper omits. |

Key EDA findings and how each changes the pipeline: (1) right-skewed series with rare spikes means MAE should be reported next to RMSE and error should be checked on high-AQI days; (2) yearly cycles mean a seasonal ARIMA or seasonal features should be tried; (3) the downward drift and shifts

after 2018 mean a walk-forward evaluation is safer than one split; (4) the ceiling near 500 means values are censored and one spike dominates min-max scaling.

## Preprocessing

The stated preprocessing pipeline is minimal: the data are split into 80% training and 20% testing samples, then mapped into the (0, 1] range using min-max normalisation (Eq. 22 in the paper) to speed up network training. After prediction, values are inverse-transformed (Eq. 23) back to the original AQI scale before RMSE, MAE and $R^2$ are computed. For the comparison models CEEMDAN-LSTM and CEEMDAN-CNN-LSTM, an additional decomposition step breaks the raw series into 9–10 intrinsic mode functions plus a residual before each component is modelled separately.

## Missing VALUES

The paper does not describe any procedure for detecting or handling missing values. Over a continuous seven-year daily collection period, gaps or invalid sensor readings are plausible, yet no interpolation, forward-filling, or exclusion criteria are mentioned anywhere in the methods section. This is a genuine omission in the reported pipeline: without knowing whether the four series are complete or how any gaps were resolved, it is not possible to assess how representative the ~2,600-point series actually are of the full study period.

## Feature scaling

Feature scaling is handled through simple min-max normalisation to (0, 1]. Because AQI is a single scalar feature, no cross-feature scaling issues (e.g. differing units or magnitudes between features) arise, which is one advantage of the paper's minimal, univariate feature set. However, the paper does not state whether the minimum and maximum used for scaling were computed from the training split alone or from the full dataset (train and test combined) before the split. If the latter, the scaling parameters would have been influenced by test-set values, a mild form of data leakage that is common in time-series preprocessing but is not addressed or ruled out in the text.

## Class imbalance

Strictly speaking, class imbalance does not apply here, since this is a regression problem with no discrete class labels. That said, the raw AQI series plotted in Fig. 5 are visibly skewed: the large majority of daily readings sit in the low-to-moderate range, punctuated by comparatively rare, sharp spikes into “unhealthy” or “hazardous” territory (e.g. AQI > 300). This is the regression equivalent of a class-imbalance problem — the extreme values that matter most for public-health warnings are a small minority of the data — yet the authors apply no weighting, stratified sampling, or peak-focused evaluation metric to compensate. Because RMSE and MAE are averaged across the whole test set, both metrics are dominated by the abundant low/medium AQI days, and the paper's otherwise strong headline results ($R^2$ up to 0.989) do not, on their own, confirm that the model forecasts rare extreme-pollution events accurately.

## Outliers, duplicates and inconsistent entries

An IQR or z-score rule would flag the pollution spikes as outliers, but these are genuine events and are exactly what the model is meant to forecast, so they should be kept (at most capped during training) rather than removed. The paper is silent on this. It also does not report checks for duplicate dates, gaps in the daily index, impossible values (for example AQI above 500 or below 0) or inconsistent definitions across cities. Possible leakage points are collected in the preprocessing audit in the next section.

## Evaluation method and preprocessing audit

**Why the paper's evaluation is not sufficient.** RMSE, MAE and $R^2$ suit regression, but the paper uses one chronological 80/20 split, one run per model, no seeds, no persistence baseline, no scale-free metric (MAPE or MASE), no significance test and no peak-day analysis. $R^2$ is also flattering on a seasonal series.

**Correct evaluation method.** Because the data are time-ordered, random k-fold cross-validation is invalid. The correct approach is walk-forward (expanding-window) validation with a separate validation segment for tuning and a test segment used once (Cerqueira et al., 2020; Hyndman & Athanasopoulos, 2021). Report MAE, RMSE and MASE as mean ± SD across folds and several random seeds, compare against persistence, ARIMA and Random Forest with a Diebold–Mariano test, and add error on high-AQI days.

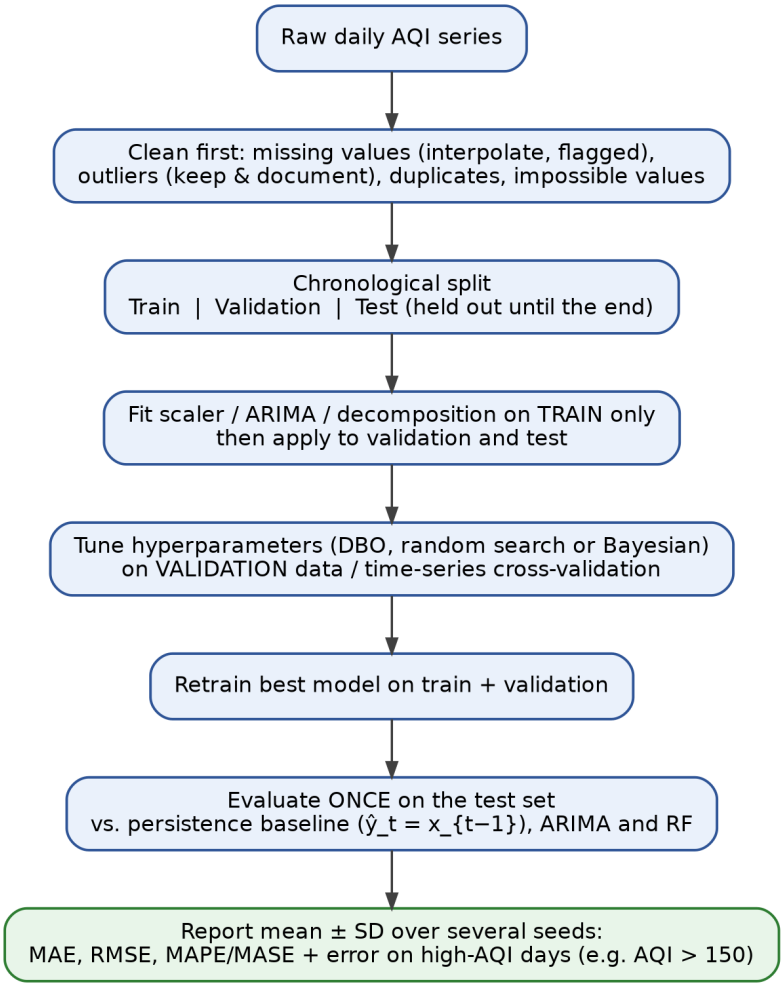



*Figure 5: Recommended evaluation workflow for this problem*

**Hyperparameter tuning strategy.** Grid search is infeasible in a nine-dimensional space where each evaluation trains a neural network. Random search is a strong, simple baseline (Bergstra & Bengio, 2012), and Bayesian optimisation is the most sample-efficient when evaluations are costly. DBO is a legitimate alternative, but it should be compared with random or Bayesian search under the same budget and scored on validation data, with nested or time-series cross-validation for an unbiased final estimate. The search spaces in Table 2 are very wide (for example neurons 1–300, batch size 1–300, learning rate 0.001–0.99), which wastes evaluations on poor settings.

**How Random Forest works for this problem.** Each row is a window of 10 lagged AQI values with next-day AQI as the target. Each tree is trained on a bootstrap sample of rows and, at every split, considers only a random subset of the 10 lags (max_features), which decorrelates the trees. Splits are chosen to reduce variance (MSE). The forest prediction is the average of the tree outputs, and out-of-bag rows give a built-in error estimate, although for time series this is optimistic because bootstrapping ignores order. Feature importance would most likely show the most recent lags dominating. A key limit is that a tree can only output averages of training targets, so a forest cannot predict above the training maximum: in Figs. 7b–10b the RF predictions flatten near 200–250 while observed AQI reaches about 450–480, which helps explain its low $R^2$ (Breiman, 2001).

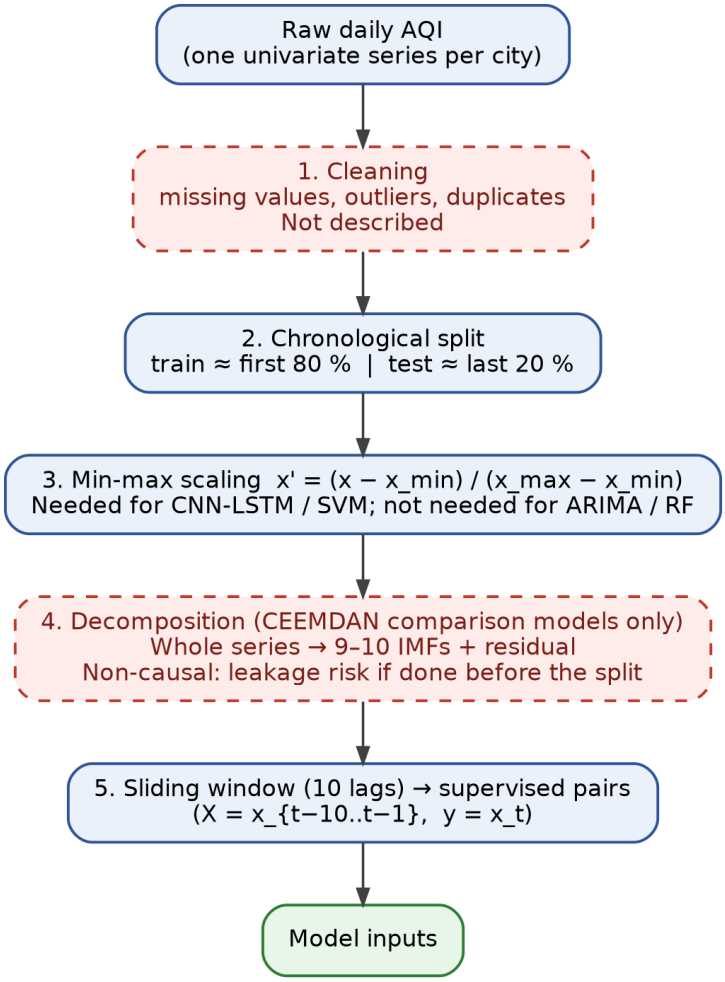



*Figure 2: Preprocessing workflow of the reviewed study*

## Preprocessing audit: where scaling and fitting could go wrong

| Check | What the paper says | Problem | Fix |
|---|---|---|---|
| Scaling (Eq. 22) | Min-max to (0, 1]; not stated whether before or after the split. | If min and max use all data, test values influence the scaler. | Fit on training data only. |
| CEEMDAN decomposition | Applied to the "raw AQI data" (Fig. 3). | Non-causal: each component uses the whole signal, so future values leak into training. | Decompose on training data only, or use a causal decomposition. |
| ARIMA residual series | Linear part fitted, residual passed to CNN-LSTM. | If ARIMA is fitted on the full series, residuals contain test information. | Fit ARIMA on the training window; roll forward. |
| DBO fitness data | Not stated. | If the test set scores fitness, tuning uses test data. | Use a validation segment. |
| Validation set | Only train and test exist. | No unbiased place to tune or stop early. | Train / validation / test. |
| Baseline tuning | Baselines use fixed hand-set values; only the proposed model is searched. | Unequal effort inflates the gap. | Give every model an equal tuning budget. |

| Check | What the paper says | Problem | Fix |
|---|---|---|---|
| Forecast mode | Not stated. | ARIMA scored on a multi-step forecast but hybrids on one-step would be unfair. | State the horizon and score all models identically. |

**Algorithm selection check.** With a single feature and about 2,000 training days, a model with roughly $10^5$ parameters is large for the data, and simpler models (SARIMA, gradient boosting on lagged values, a GRU) might perform similarly with lower cost. A jump from $R^2$ of about 0.27–0.5 to 0.96–0.99 by adding a linear model to a neural network, and the almost perfect diagonal in the scatter plots of Figs. 7b–10b, are unusual for daily pollution data and are consistent with information leakage. This should be tested in Phase 2 by re-implementing the hybrid with strictly train-only fitting and comparing it with a persistence baseline, a Random Forest and one alternative model.

## Summary

Duan et al. (2023) frame AQI forecasting as supervised univariate time-series regression and combine ARIMA for the linear part, a DBO-tuned CNN-LSTM for the non-linear part, and additive recombination. Their ablation design supports combining the components, but the evidence is weakened by unreported missing-value handling, an unclear order of scaling and decomposition relative to the split, no validation set or stated fitness data for DBO, baselines tuned unequally and lacking a persistence benchmark, a single run on a single split, and numerical inconsistencies in the text and figures. The reported $R^2$ of 0.95–0.99 therefore cannot yet be accepted as genuine forecasting skill. Phase 2 will test this with a baseline implemented under a time-based split and train-only fitting, and Phase 3 will build the full critique and proposed alternatives on these findings.